LESSON 8.3

Write and Solve Equations Using Multiplication or Division

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve one-step multiplication and division equations using inverse operations.

WHY THIS MATTERS Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

REMEMBER

Write and Solve Equations Using Multiplication or Division

Solving Multiplication & Division Equations. Formula: ax = b → x = b/a | x/a = b → x = a · b When a variable is multiplied or divided by a number, I use the inverse operation to get it alone. Division undoes multiplication, and multiplication undoes division.

1. Identify the operation applied to the variable.
2. Use the inverse operation: Divide to undo multiplication; multiply to undo division.
3. Balance: Apply the inverse operation to both sides to isolate the variable

Words: **Inverse operation** ***(Operación inversa)*** **Multiply** ***(Multiplicar)*** **Divide** ***(Dividir)*** **Isolate** ***(Despejar)*** **Coefficient** ***(Coeficiente)***

1 **Circle the letter of the best answer. Show how you know.**

Solve: m − 11 = 25

(A) m = 14
(B) m = 25
(C) m = 36
(D) m = 275

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Solve: n + 15 = 42

(A) n = 15
(B) n = 27
(C) n = 42
(D) n = 57

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: y − 7 = 20

(A) y = 7
(B) y = 13
(C) y = 27
(D) y = 140

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Solve: x + 9 = 15

(A) x = 6
(B) x = 9
(C) x = 15
(D) x = 24

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A detective ordered fingerprint kits for her team. Each kit costs $7, and the total bill was $63. The equation 7k = 63 represents this situation, where k is the number of kits. Solve 7k = 63.

(A) k = 9
(B) k = 56
(C) k = 70
(D) k = 441

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A detective ordered fingerprint kits for her team. Each kit costs $7, and the total bill was $63. The equation 7k = 63 represents this situation, where k is the number of kits. Which operation undoes multiplication?

(A) Division
(B) Addition
(C) Subtraction
(D) Multiplication

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide to solve an equation?

SENTENCE STARTER When I see ___ in the equation, I know to ___ both sides because the inverse of ___ is ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can solve one-step multiplication and division equations using inverse operations.

WHY THIS MATTERS Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Level 2 Extension - Match each locked vault to the solution that opens it.

_____ 7x = 56

_____ x / 5 = 4

_____ 11m = 33

_____ p / 2 = 9

(A) x = 8

(B) x = 20

(C) m = 3

(D) p = 18

2 **Write the name of the correct group on each line.**

Detective, decide which key opens each vault: do you MULTIPLY or DIVIDE to unlock the variable?

Groups: Divide both sides | Multiply both sides

_____ 6x = 48

_____ 9k = 27

_____ x / 8 = 2

_____ n / 4 = 5

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A detective ordered fingerprint kits for her team. Each kit costs $7, and the total bill was $63. The equation 7k = 63 represents this situation, where k is the number of kits. Check: does 9 work?

(A) Yes — 7 × 9 = 63

(B) No — 7 × 9 = 56

(C) No — 7 + 9 = 16

(D) Only if kits cost $9

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Vault A is sealed with 4x = 32. Crack the lock: what is x?

(A) x = 8

(B) x = 28

(C) x = 36

(D) x = 128

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Vault B is sealed with x / 3 = 6. Crack the lock: what is x?

(A) x = 2

(B) x = 3

(C) x = 9

(D) x = 18

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Vault C is sealed with 5n = 45. Crack the lock: what is n?

(A) n = 9

(B) n = 40

(C) n = 50

(D) n = 225

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide to solve an equation?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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